Problem Statement

Diagnosing repeat purchase decline for a UK online gift retailer

The business believes fewer customers are coming back for a second order, but nobody can say who is leaving, when they leave, or what it is costing. This project sets out to answer that, using two years of real transaction data from a UK-based online retailer of all-occasion giftware, and to end with a recommendation the business can actually act on.

1.05M	5,852	24 MO	40+
RAW LINE ITEMS	IDENTIFIED CUSTOMERS	DEC 2009 TO DEC 2011	COUNTRIES SOLD TO

THE SITUATION

The retailer sells all-occasion giftware online, with a customer base that mixes casual, one-off gift buyers and wholesale accounts placing large repeat orders. There is no subscription and no cancellation button, so unlike a subscription business, nobody tells the company when they have decided to stop buying. A customer simply stops ordering, and the business is often the last to notice.

That makes the underlying question harder than it first sounds. A drop in repeat purchases could mean the product is not landing, or it could just as easily mean a strong holiday season pulled a lot of one-time buyers through the door who were never likely to return in the first place. Without breaking the data down by who is ordering, when they first showed up, and how their behavior changes afterward, those two explanations look identical from the outside.

WHY IT MATTERS

Acquiring a new customer costs more than keeping one, and a retailer that cannot tell who is at risk of leaving has no way to intervene before it is too late. If the business is quietly losing a specific, identifiable group of customers, whether that is a certain type of buyer, a particular time of year, or a specific stage in their relationship with the company, that is something a targeted campaign can fix. A vague sense that "retention feels soft" is not.

WHAT THIS ANALYSIS NEEDS TO ANSWER

- Who counts as a lost customer in a business with no formal cancellation, and how long of a silence is long enough to call it churn.
- Which customers are already gone, which are drifting, and which are still healthy, in terms specific enough that someone could act on the list today.
- Whether newer customers are being retained better or worse than customers who joined earlier, and at what point in a customer's first few months the relationship is most likely to break.
- What a churned customer is actually worth, so any fix can be weighed against what it would cost to run.

SCOPE AND HONESTY ABOUT THE DATA

About a fifth of all orders have no customer identifier attached, meaning they cannot be tied to an individual over time. Those are kept for revenue and product-level questions but left out of anything that tracks a specific customer, so the retention numbers in this project describe about 87% of revenue, the share that can actually be traced to a person, not the full business. That gap is itself worth a mention to the stakeholder, since it points at a real data-capture problem at checkout, not just an analysis limitation.